

IMPERIAL COLLEGE OF SCIENCE, TECHNOLOGY AND MEDICINE

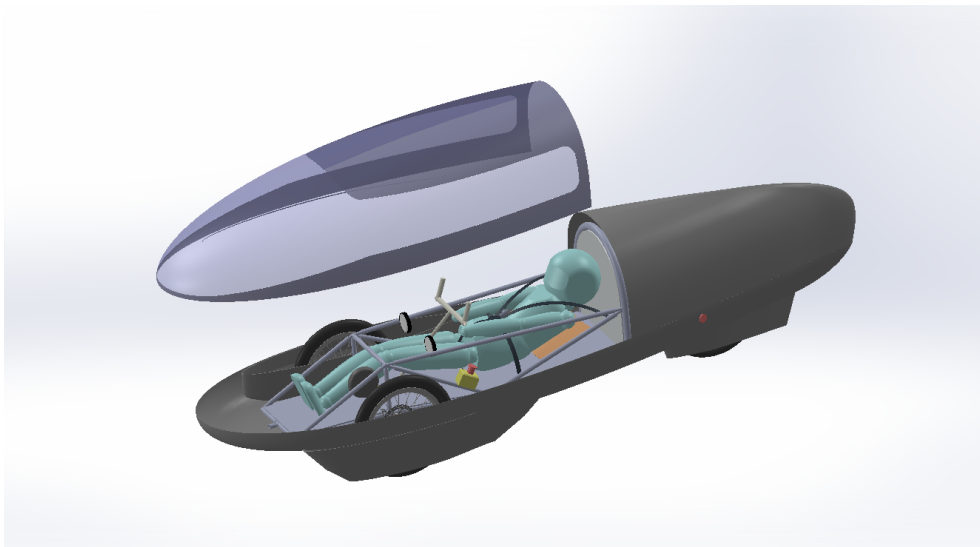
DEPARTMENT OF AERONAUTICS

MENG AERONAUTICAL ENGINEERING

GROUP DESIGN PROJECT

Shell Eco-Marathon

Rules and Regulations Compliance



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Abstract

The objective of the group design project was to design a prototype electric battery vehicle to compete and ultimately win the Shell Eco-Marathon competition at Mercedes-Benz World in July 2020 (note due to Covid-19 this event has been indefinitely cancelled). The purpose of this report is to demonstrate that the vehicle designed conforms to the rules and regulations of both the external stakeholder i.e Shell and also the internal stakeholder i.e Imperial Racing Green. The report details the objectives of the compliance team and provides a reflection at the end to discuss the positive impact that the compliance team had on the success of the project. It also demonstrates how changes made to the design were managed across sub-teams to ensure no rules or regulations were subsequently breached. Ultimately the car designed was deemed fully compliant to all of the rules and regulations.

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1 Executive Summary

1.1 Technical Director and Project Coordination

The objective of the Shell Eco-Marathon Group Design Project was to design a prototype electric battery vehicle to compete and ultimately win the Shell Eco-Marathon competition at Mercedes-Benz World racetrack in July 2020, with a budget of £8000. Even though the competition for 2020 has been cancelled, this design project provides IRG a vehicle platform that can be built upon for the next round of competitions (summer of 2021 as of now). The proposed design has two wheels at the front and one at the rear, to accommodate a tear-drop shape low drag aeroshell. The vehicle has Front-wheel steering per the SEM regulations and a rear-wheel drive. The vehicle's power plant is a 180W Brushless Maxon motor, powered by 8 Samsung INR18650-25R cells. The vehicle has a space frame style CFRP chassis and an integrated aluminium 2090 T83 roll bar. The aeroshell is a non-load bearing structure made up of custom top, side, and under-body profiles with integrated wheel fairings. It is constructed from CFRP and a transparent Polycarbonate canopy to maximise driver visibility. The teams choice of tyre for the design was the Michelin 4575R16, these offer the lowest rolling resistance coefficient of any tyre available on the market and are manufactured specifically for this competition. The final mass estimation without the driver is 25.49 kg. The vehicle was designed around the selection of a 10th percentile female driver, of mass 50 kg when wearing full race gear.

A performance metric was constructed to analyse the sensitivity of the vehicle performance to varying key parameters, with a final performance estimation of 1141 km/kWh. This performance estimate exceeds our target of 900 km/kWh. We acknowledge that our performance metric only models the ideal conditions and does not consider losses due to driver error. This design project was solely based on theoretical work and for future adaptation, we propose experimental testing of individual components to IRG. Based on the sensitivity analysis of the final design the team believes the vehicle's aerodynamics is the area where the largest performance gains could be made. As such we advise IRG to take advantage of our vehicle's modular design which allows the aeroshell to be easily replaced without requiring a large redesign of the vehicle to build and rigorously test different aeroshells.

Please refer to the report by Davide Vaccaro for detailed information on 'Technical direction and project coordination'.

1.2 Rules and Regulations Compliance

Without appropriate regulation compliance the design could be disqualified from the competition at the technical inspection. The purpose of the rules and regulation compliance team was to ensure the vehicle produced is fully compliant with the rules and regulations set out by the organiser, Shell, and also compliant with the IRG internal rules surrounding budget and other constraints e.g. driver dimensions. The rules and regulations for Shell are documented in 2 rule books which can be found online. By populating a spreadsheet the rules could be broken down between the sub-teams and managed accordingly.

The first focus for the compliance team were the conceptual design regulations, namely the dimensions. Once this was completed the focus shifted onto the sub-team specific rules. These were regularly checked for compliance as the design progressed.

Any discrepancies with the design or unclear rules were managed appropriately by discussing the problem with Imperial Racing Green and coming to a joint conclusion. The vehicle was deemed to be fully compliant by the end of the design process.

Please refer to the report by Emily Middlebrough for detailed information on 'Rules and Regulations Compliance'.

1.3 Packaging

Integration of different sub components is an essential aspect for the success of the project and it is the role of Packaging to communicate efficiently between different sub-teams. The integration of the car relied upon a constantly active line of communication between each sub-team and the packaging team. When a change was made by any sub-team, to ensure the smooth running of the project, the packaging team were required to use their 'big-picture' view of the project to quickly and efficiently communicate any knock-on effects to other sub-teams.

At the start of the project the team helped with the outlining of the conceptual design by considering the possible configurations of car. The packaging team helped the initial decision to go with a three-wheel configuration with two front wheels, a single rear wheel, front wheel steering and rear wheel drive. As the project developed the packaging team worked closely with the different sub teams to assist them in any major design decisions, such as the steering system, power train mounting and window sizing. In order to make any changes clear to the whole team, a SolidWorks drawing was made easily available for any team member to reference and gather any required dimensions. A centre of gravity chart and a version control system documenting different versions of design is also maintained throughout the project.

Please refer to the reports by David Simmonds and Zijun Liao for detailed information on ‘Packaging and Integration’.

1.4 Aerodynamics

The Aerodynamics team had the task of designing an aerodynamically competitive shell whilst making sure to comply with race regulations and the vehicle’s packaging constraints. For this reason, much time was spent working with sub-teams in charge of the chassis, wheels, steering, packaging and driver environment to produce a vehicle concept that would allow a major aerodynamic improvement compared to the current IRG design. With much inspiration taken from 2 world record holding contenders, a vehicle with a relatively high ground clearance and most importantly, wheel fairings, was produced. In addition to a custom top view “teardrop” airfoil shape and long slender profiles for the side view and wheel fairings, the final vehicle showed much better drag characteristics than the previous IRG design. The shell is also expected to improve the visibility issues previously experienced, with the whole front half of the car’s canopy made fully out of transparent polycarbonate material using only tinting to reduce direct sunlight on the driver and the remainder of the shell produced out of carbon fiber reinforced epoxy resin.

The final drag coefficient is estimated to be around 0.075 deemed to be highly competitive amongst previous top contenders. However, there is still work to be done in the future including a reworked fairing and under-body integration to ensure neutral lift, improved CAD to allow tighter packaging around the roll bar and wheels for a smaller frontal area and finally, wind tunnel testing to validate the results seen in CFD simulations. *Please refer to the reports by Toby Bryce-Smith, Xerxes Chong, Cedric van Den Berg and Michal Wietlicki for detailed information on ‘Aerodynamics’.*

1.5 Drive Train

The drive train includes motor, transmission, tyre and wheel. Motor selection began with power requirement estimation using initial sizing data provided by Packaging and Aero Teams. A preliminary mission profile was designed with help from Vehicle Dynamics Team. An algorithm was developed with MATLAB which utilised technical data from manufacturer to compute energy usage for shortlisted motors during the race. Through comparison of mileage, mass and optimal gear ratio, Maxon 574741 was selected. The predicted mileage using primary mission profile was 1347.5km/kWh, with optimal gear ratio of 4.3. Motor data was sent to Electronics and Vehicle Dynamics Teams for race strategy analysis. Final mileage achieved was 1141.15 km/kWh.

For transmission, shaft design was chosen, as it was light and offered very high power transmission efficiency of 99%. Given optimal gear ratio of 4.3 no planetary gear or multi-step transmissions were required. The final design consisted of a woven CFRP shaft perpendicular to the rear axle connected to the motor, which would be mounted in between two chassis members at the rear of the car. Translation of the torque would be achieved by a straight bevel gear. The bevel gear was mounted to a standard bike freewheel cassette to allow the wheel to coast without overloading the motor. A frame was designed to ensure alignment of the axles, and smooth meshing of gear teeth.

Michelin Prototype 45/75R16 tyre was chosen as it had the lowest rolling resistance coefficient of 0.00081 in all rubber-made tyres. For wheel design, customization of BMX wheel was chosen due to its high cost-effectiveness. Customized 20” Alienation TCS VANDAL rim was chosen to firmly accommodate the tyre. Shimano RS770 hubs were chosen for brake disc and gear compatibility. Other wheel parts would be purchased separately and assembled with the help of truing stand in the bike storage in the college.

Please refer to the reports by Pengchao Hu and Lorenzo Tresca for detailed information on ‘Drive Train’.

1.6 Vehicle Chassis

A spaceframe chassis built upon a ladder chassis base was designed to support the driver and the components of the chassis. This is primarily based on the *Vehicle Dynamics, Driver Environment and Packaging* constraints and allows flexibility for any dimensional changes during the conceptual phase. As the design progressed, the space frame adaptation approach would be used.

The Engineer's Theory of Bending is used to analyse the deflections of the chassis base during the conceptual design phase. The cross section of the structure is chosen based on the relations between torsional loading and twist. A truss structure analysis was done to identify the optimum chassis sides and verify the effectiveness of its implementation on reducing the deflections. As the chassis structure increased in complexity, analysis was done purely based upon FEA results. Additional members were included and the diameters of the members under critical stresses and deflections were increased to improve the performance of the structure.

The final chassis structure gives a maximum deflection of 0.69 mm under static loading and a 3.97 mm maximum deflection and 2.09° rotation in the worst case of dynamic loading, which are within the requirements of the IRG team. The spaceframe will be made of roll wrapped carbon fiber tubes, assembled using Al 6082 inserts. The tubular members and the inserts will be bonded together with HexBond 870 Shimming Epoxy. The final mass of the assembled space frame is 3.24 kg and the estimated cost is £534.

Please refer to the reports by Samuel Cross, Sakishna Linganathan, Angeliki Tzola and Imran Ahman Azhar for detailed information on the 'Vehicle Chassis'.

1.7 Loading and Crash Protection

The aim was to design crash safety structures to protect the driver from possible impact conditions of the vehicle. The lateral instability of the vehicle made roll overs the most probable form of impact. Regulations required a roll bar or panel to be constructed which could withstand a load of 700 N without significant deformation.

Industry grade techniques of pre-processing, simulations and post-processing were adopted to design an efficient Roll bar. The pre-processing stage included thorough research in material and cross-section selection, possible joining methods, adoption of efficient 2D finite element methods to choose the optimum shape and lastly, to obtain the range of possible deflections due to different degrees of clamping. The post-processing stage came after the 3D simulations where detailed conclusions were made with respect to the complete failure analysis of the roll bar, accuracy of the 2D simplified model and the robustness of the clamping assumptions made in the 2D analysis. These steps concluded a maximum displacement of 3.8 mm with the limit set at 5 mm.

Furthermore, this process assured crash protection in case of roll overs.

Other possible impact conditions were assessed i.e. driver entry and crumple zone regulations. Overall, the vehicle was concluded to be safe in case of probable crashes and possible improvements were listed to further optimise the structures.

Please refer to the report by Devanshu Ghosh for detailed information on 'Loading and Crash Protection'.

1.8 Power and Control

The role of the Power Electronics and Control sub-team was to simulate performance of power systems and work with Powertrain and Vehicle Dynamics sub-teams to optimize component and strategy selection respectively. A model for all electronic systems in the powertrain was therefore developed and integrated with a dynamics model to simulate the vehicle's performance throughout the race. In particular, the performance and integration of the motor controller was analyzed.

Particular focus was also placed on driver interaction with the vehicle's power systems. An intuitive control scheme for the vehicle's cruise control was created in order to maintain a constant speed throughout the race and improve efficiency. At request of IRG, a real-time telemetry system was also designed that would transmit sensor data from the vehicle to the team using cellular networks while serving as a central vehicle computer with control over throttle actuation, driver displays, and the aforementioned cruise control system.

Finally, the Power Electronics and control sub-team was responsible for maintaining a list of project deliverables and consulting any suggested changes with IRG to ensure their needs, as our customer, were met.

Please refer to the report by Tomas Mrazek for detailed information on 'Power and Control'.

1.9 Braking System

Theoretically, if the driver was to follow the race strategy, brakes will not be used during the race. However, the brakes still play a big part in the competition as there will be a technical inspection test which the brakes are required to pass whilst complying with all the rules. In order to pass the braking test, the vehicle - with the driver inside – placed on a slope with an inclination of 11.3° , must remain stationary. Nevertheless, the brakes have an important role for safety and efficiency purposes.

Considering factors like speed, weight and tyre size of the brakes, a brake similar to a mountain bike brake is chosen due to the similarity in the aspects listed earlier. The selection of brake components from the market was based mostly on weight reduction and performance; in order to increase the overall performance of the car. The merits of typical bicycle brakes are assessed and Shimano BR-M8110 disc brake callipers and BL-M8100 hand brake lever are used. Meanwhile, the rotor disc will be made in house by IRG and the optimised foot pedal will be 3D printed in house using ABS due to its lower weight and price considerations. Braking power is transmitted by Shimano SM-BH90-SS Brake Hose which is routed along the chassis and steering column. The front callipers will be side mounted on the optimised brake arm and the rear brake calliper will be flat mounted like a bike. The handbrake lever will be mounted in the middle of the vertical handlebar adaptor. Overall, the performance of brake system would also depend on the installation quality and the actual brake test.

Please refer to the reports by Pok Wei Hui for detailed information on ‘Braking system’.

1.10 Suspension and Steering

Suspension was decided to not be implemented in our vehicle design, as no stability issues were experienced during the required race manoeuvres as calculated by *Vehicle Dynamics*, and potential contact of wheel fairings with the ground due to wheel travel.

The vehicle has 2 front and 1 rear wheel, and the rules dictate front wheel steering only is permitted. It is operated with a handlebar with vertical attachments at the hands for more ergonomic use, and utilising a universal joint to transmit rotation to a pitman arm. The pitman arm and tie rods operate under the chassis floor, as this is safer and less invasive to the driver with no moving connections in their compartment.

Full Ackermann steering was used to allow appropriate independent inner and outer wheel turn angles to prevent the wheels fighting and slipping. The max wheel turn angle is for the 8m radius turn test at technical inspection, a safety margin of 20% was added as our full lock angle to allow for any unpredicted behaviour of the wheels and for any mismanufacture of components and the car. The steering was modelled to have a minimum steering ratio of 1:1 for the inner wheel to provide controlled and responsive steering, whilst allowing a non-invasive turning of the handlebar and the driver.

Please refer to the report by Corey Taylor for detailed information on ‘Steering and Suspension’.

1.11 Driver Environment and Interfaces

The first task for the Driver Environment and Interfaces team was to select a suitable driver and optimise the seating position. The most important aspect for the driver was minimising weight and so a female in the lowest 10th percentile weight category was chosen. A supine seating position was selected due to the corresponding decrease in the frontal and wetted areas of the vehicle.

The next stage was to consider the driver interfaces. The steering wheel, being the centre of the interface, was designed to be a compact handle bar to maximise comfort and minimise weight. It was packaged with the break lever, horn button, speedometer and throttle. The transparent polycarbonate aeroshell and a pair of rear-view mirrors were put in place to ensure the visibility requirements were met.

The final task was to consider the safety and comfort of the driver. A 5-point harness was integrated with the chassis to ensure proper load distribution and a horn was added to avoid collisions. The team then worked alongside the Chassis team in order to combine the side supports of the chassis with the seat. A ventilation system was designed in conjunction with the visibility holes for the horn and a hydration system was also included.

Please refer to the reports by Matilda Cole and Tianyi Sun for detailed information on ‘Driver Environment and Interfaces’.

1.12 Vehicle Dynamics and Traction

The role of the Vehicle Dynamics subgroup in this project was to identify how the changes in various aspects of the design affect vehicle performance and stability. The main aspects were focused on include: total mass of the vehicle with the driver, drive train parameters, aerodynamic characteristics, tyre selection and tyre forces. The initial tyre force model was created by estimating the minimum cruising speed required for the car to complete the race within the time limit set. The initial tyre force estimations were then used to validate the initial vehicle dimensions proposed by the Chassis team and it also allowed the Steering and Drive train teams to have an estimate of what the maximum required steering angle and traction force would be. Once the Chassis team was close to finalizing the vehicle dimensions, an FEA analysis was then performed using the maximum tyre loads obtained from the tyre model. A lap simulator was also created and validated in order to predict the vehicle performance based on energy consumption of the car throughout the race, which motivated the study into energy efficient driving strategies once the basic accelerating then cruising velocity profile was created as a baseline for comparison. The lap simulator script which was written in MATLAB was later put together with the other sub-teams' scripts which output variables such as the aerodynamic and tyre forces on the car, motor power and battery simulation. Stability checks on steering, tyre grip and rollover were also included within the lap simulator to ensure final vehicle stability.

Please refer to the reports by Qirui Wang and Nicholas Tham for detailed information on 'Vehicle Dynamics and Traction'.

1.13 Battery and Cooling

The role of the Battery and Cooling sub-team was to design an electrical energy storage unit capable of storing and releasing the required energy during the race; this was found to be 35 Wh, with an output-voltage > 15 V and current up to 10 A. The use of Lithium-ion batteries, capacitors and structural batteries were considered; however, latter was ruled out by the regulations and it was found that using Lithium-ion batteries provided the lightest solution. A circuit model was produced to model the battery discharge which was incorporated the drivetrain and motor controller. A battery management system was configured and metal tray incorporated into the casing to meet the required regulations.

Final estimation of the power consumption was 15 Wh. Hence, a smaller battery would suffice, now limited by the voltage requirement. This would result in a 120 g weight reduction. However, this reduction can also be realised, without capacity loss, by using higher capacity cells at the additional cost of £27. IRG required the vehicle temperature to remain below 40°C. To estimate the vehicle temperature the heat transfer modes were modelled in the worst-case scenario (no wind, $T_a = 30^\circ\text{C}$, $T_{track} = 60^\circ\text{C}$). The main heating effects were found to be solar gain and driver heat. To mitigate these effects a reflective film covering with low absorbance was used and a ventilation inlet was sized at 1600 mm². Future recommendations include decreasing the absorbance by tinting of and/or reducing the size of the window as well as looking into the possibility of making the driver more comfortable at hotter temperatures allowing higher temperatures, resulting in a reduction of the inlet size and thus drag.

Please refer to the report by Gregorius Campman for detailed information on 'Battery and Cooling'.

1.14 CAD & Integration

The role of CAD & Integration team is to build a CAD model of the car and update it weekly. The CAD models were generated in Solidworks 2019 and were used to visualize each component, check the compatibility of each component, optimize space and calculate the center of gravity. This ensures the chassis would not stick out of the shell, front wheels would not touch anything while turning and enough space was designed for the driver to comply with the regulations, etc.

Since a CAD model is essential for CFD analysis, we work closely with *Aerodynamics Team* to keep updating the model of the shell. We started from a rough design with only top and side view to a complete aeroshell design with wheel fairings. Also, we work closely with *Chassis Team* to update the model from a simple plate to a 3D space frame to give the team a general concept of the car at every stage. By working with *Steering, Braking, Drive Train, Battery and Driver Interference Team*, all components were packed within the confines of the shell.

Please refer to the reports by Yuheng Wang and Yueqing Yang for detailed information on 'CAD & Integration'.

2 Introduction

If the car designed fails to be compliant with any of the rules set out by Shell, it will suffer penalties and potentially disqualification for more serious infractions. The rules and regulations for the competition are exhaustive – a 47 page rule book is provided by Shell [1] concerning the design constraints as well as a further 13 page rule book specific to the race day constraints. There are specific rules for urban concept and prototype vehicles and as this design was prototype the urban concept rules could be discarded. Additionally, further constraints are set by Imperial Racing Green in terms of the budget available which is around £12000. The purpose of these rules is primarily to ensure safety for not only the driver, but also spectators, race officials and other drivers on the track. Without a structurally sound set of rules and regulations, the competition would not be allowed to legally take place, hence Shell take compliance to these rules seriously and will disqualify any team found to invoke them.

3 Compliance Objectives

Compliance is fundamental for any business to create a legal and safe product. The approach for managing compliance in this project was taken as follows:

- Accurately record all known rules and regulations from Shell and Imperial Racing Green
- Have a designated compliance team
- Make all sub-teams aware of the constraints affecting them
- Ensure effective methods of communication when making changes
- Make all teams aware of the consequences for not being compliant
- Immediate flagging of any concerns and discuss with relevant parties
- Keep project manager aware of compliance evolution throughout the project

4 Compliance Team Integration

The compliance team was formed of one individual and supported each of the 5 sub-teams while reporting to the project lead. The whole team was split into several multi-disciplinary sub-teams: body, power, packaging and integration and dynamics. Compliance sat across all of these teams to ensure any changes made were not detrimental to not only that teams' regulations, but also any other affected teams. This also enabled a whole view perspective to be held by the compliance team and allowed it to become actively involved in design decisions across the project. A diagram of the project organisation can be found in Appendix A.

5 Compliance Management

Due to the fact it would be untimely for each team to constantly be searching for rules which applied to them, it was decided upon for a spreadsheet to be created in which every rule both internal and external was documented. Each rule was assigned a primary sub-team i.e the sub-team which was most affected by this rule, for example, the battery rules were assigned to the power team. Following this initial sorting, each rule then had any additional sub-teams noted that would also be potentially affected by a change in that rule compliance, for example, the safety harness could affect the chassis design. Finally, the rules were highlighted either red, orange or yellow and this signified:

- **Red – Critical rule**
 - Vital for the design to continue in a compliant manner
 - Would result in major whole design changes if found to be non-compliant
- **Orange – Be aware rule**
 - Generally more sub-team specific so less impact on others' designs
 - Would generally only result in sub-team design changes if found to be non-compliant
- **Yellow – Future issue Rule**
 - Not relevant for the scope of this project
 - Needed by Imperial Racing Green for the next stage of the project

The proportion of rules which were deemed to sit in each of these categories are shown in the table below.

Rule Category	Percentage of all Rules
Critical	14%
Be aware	68%
Future issue	18%

Table 1: Table showing proportion of rules.

From this table it can be noted that only 14% of the rules are critical, however if any of these rules are not followed it would be extremely detrimental to the project to take time to correct these mistakes. The number of rules to be focused on for the purpose of this project excludes those deemed a future issue so the rules to focus on decreases by 18% to around 89 rules. Each rule was investigated against the existing Imperial Racing Green design to ensure we did not adopt any existing erroneous designs. No issues were found in this.

6 Compliance and Disqualification

6.1 Compliance Evolution

The compliance of the design was closely monitored throughout the project and an approach was taken to ensure the areas of focus for compliance shifted as the project progressed.

In short, the first week was spent focusing on the conceptual design and thus the rules which were focused on were the main dimensions of the car, namely the height, length and height to track width ratio.

In the subsequent weeks and as the design became more detailed, the more intricate rules were focused on for each sub-team. For example, the battery capacity and the type of safety harness required.

Compliance was in touch with every sub-team multiple times a week to discuss any design concerns and confirm compliance for what had been achieved.

Whilst some design queries could be adequately answered by compliance, others were more complex and did not have a straight forward answer. These queries were researched by compliance and then discussed with a contact from Imperial Racing Green, Sampson Tam. Once compliance and Sampson had discussed the issue in question, the information was fed back to the relevant team. This type of query took a maximum of 2 hours to solve which in turn ensured the design was running to schedule whilst also being compliant.

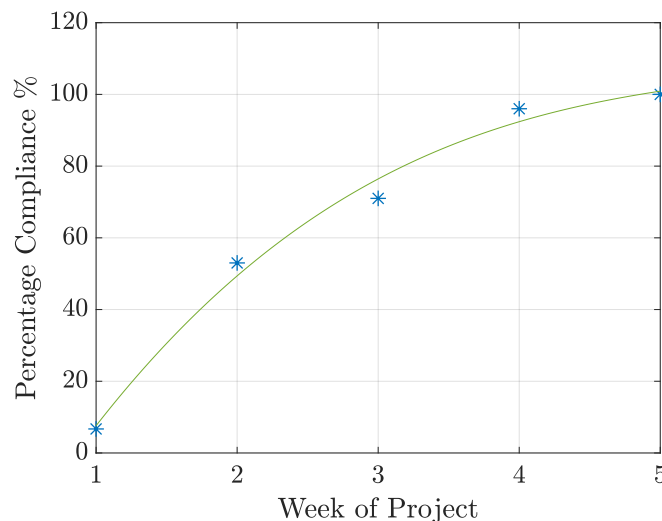


Figure 1: A graph showing how compliance progressed throughout the project. It can be seen that the compliance percentage increases at a faster rate towards the start of the project. This is expected, as the majority of the rules have to be met early on in order for the design to be iterated and still be legal. Only small changes needed to be made towards the end, hence the compliance barely altered as most rules were already being adhered to.

6.2 Disqualification

Disqualification penalties for not being compliant with any rules, both safety and technical are as follows:

- 1st Infraction – Formal warning
- 2nd Infraction – Best overall attempt invalidated at the end of the competition
- 3rd Infraction – Immediate team disqualification [1]

7 Conceptual Design Compliance

The purpose of the conceptual design was to show to main design principles of the vehicle and the main constituents that ought to be focused on in the detailed design stages. The regulations focused on for this stage in the project were wholly the dimensions which included the length, width, height and height to track width ratio. The conceptual design below shows some of the main dimensions and they conform to the length being less than ≤ 3500 mm, width less than ≤ 1300 mm, height less than 1000 mm and height to track width ratio less than 1.25.

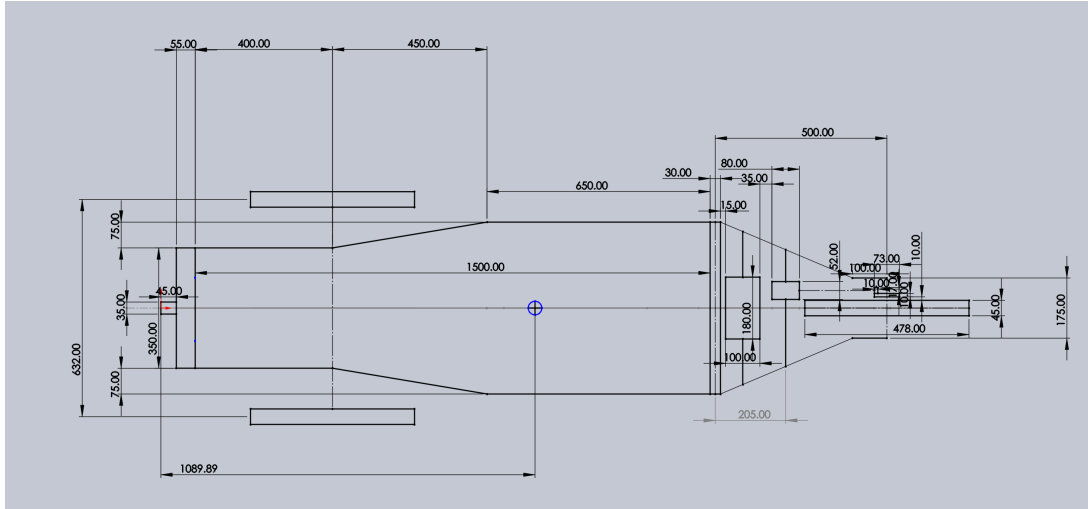


Figure 2: Drawing of the conceptual design which passed compliance tests [2].

8 Detailed Design Compliance

8.1 Aerodynamics

From the outset of the project the aerodynamics team took a technical backseat as there were other more pressing issues to be sorted before the aerodynamics shell became priority.

The main rules surrounding aerodynamics explicitly state that there should be a crumple zone ≥ 100 mm, aerodynamic appendages were prohibited, any sharp points on the aerodynamics shell had to be ≥ 50 mm or covered with foam/deform-able material and the body panels and windows must be rigid and thus not change due to wind. These were met by not having any appendages, the sharp point at the rear of the shell was increased so that it was ≥ 50 mm and the body panels and windows were chosen to be made from materials that were substantial enough to not change with wind. Note that the rules were not explicit about the strength of wind that the body ought to be able to withstand therefore an average wind speed for the location at that time of year was chosen.

The materials chosen for the aerodynamics shell to fit the strength criteria were Lexan and CFRP (exact one dependant on budget). Aerodynamics also played a role in the visibility requirement that also concerned driver environment. The constraint was that the driver has to have a direct arc of 90 degrees visibility. This was flagged as a previous issue for the Imperial Racing Green project and it has been known to see teams heavily penalised in past competitions. To counter this, the aerodynamics team discussed having an entirely transparent top. However, given the heating issues with the driver environment it was deemed too risky as it effectively would allow the driver to become excessively hot if the sun was shining. A compromise was made between compliance, aerodynamics and also driver

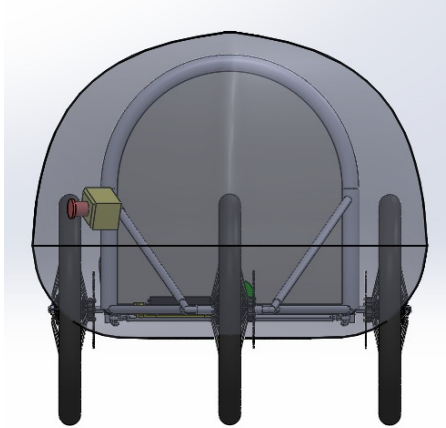


Figure 3: The aerodynamics shell from the rear view, the rounded edge can be seen from this on the center line. This CAD model is sourced from the CAD report [2].

environment, to have a Lexan front with tinting to ensure the driver is not over-heated yet visibility requirements are still met.

8.2 Battery and Cooling

The battery and cooling sub-team had a large number of regulations. There were 22 direct constraints relating to the battery size, maximum voltage and also location. As the battery is a major factor in safety issues due to overheating and fire hazards, great care was taken to ensure these requirements were met with a satisfactory safety margin.

The battery was constrained by a maximum voltage of 60 V. The battery selected achieved this constraint with a maximum voltage of 28.8 V. The management system was designed to meet the requirements of cell-balancing and over-load protection. Finally, the battery had to be located away from the driver in an enclosed environment to mitigate any risks associated with lithium battery fires. The battery was selected and calculations were made to predict the rise in temperature of the driver environment due to the presence of the battery. This was calculated to be a rise of a maximum of 14 degrees which satisfied the constraints. It was discussed that the weather on a summers day in Surrey could well be up to 30 degrees so this additional 14 degrees could make the driver overheat. Driver environment ensured this issue was mitigated against by selecting an appropriate suit to protect the driver also providing ventilation through the aero shell.

The cooling team members worked closely with compliance and also driver environment to ensure the operation of the vehicle would not be compromised by excessive temperatures. Vents were therefore discussed and it was decided that the holes which allow the horn to be seen situated on the front of the shell would also double as ventilation. Further information on this can be found in Matilda Cole's report [3].

The location of the battery was considered to meet the regulations, it should be separate from the driver's compartment and should be protected against open flames. To satisfy this, the battery was placed behind a fireproof bulkhead as shown below.

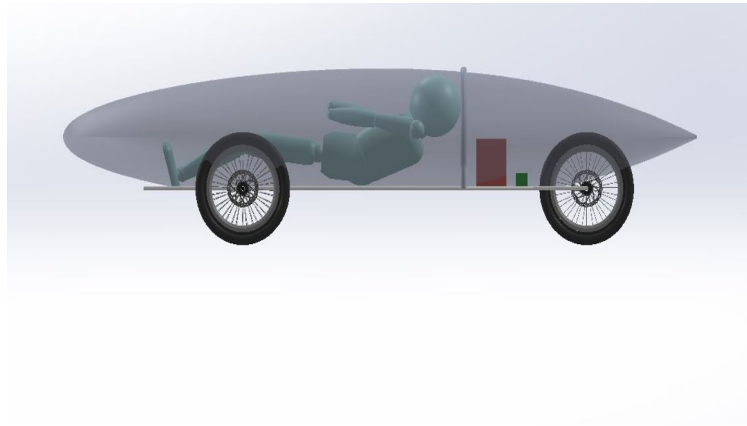


Figure 4: The location of the battery behind the bulkhead [4].

8.3 Chassis

The primary concern of the chassis is to protect the driver's body in a crash. It should be designed while adhering to the dimensional constraints set out in the first week of the project. Once the chassis was compliant with those constraints, any additional changes were to optimise the structure and provide more optimal torsional stiffness but these changes bore no effect to the compliance. The chassis evolution from the conceptual design was severe as the original had major dimensional errors as can be seen from the model below.

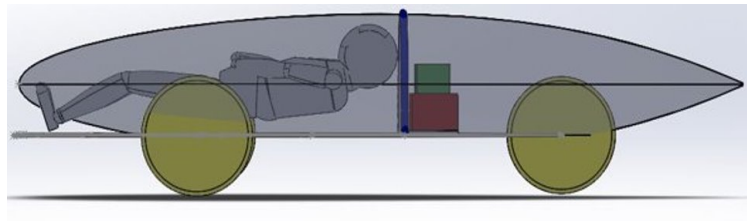


Figure 5: First chassis design, clearly non-compliant, image courtesy of the CAD team [2].

Once the dimensions were compliant and agreeable to the aerodynamics shell, the chassis was optimised for the loading expected during the event. The result was a fully compliant chassis capable of withstanding the expected torsional loads.

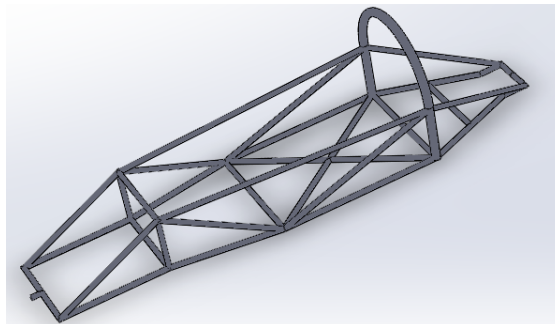


Figure 6: Final chassis design, fully compliant, image courtesy of the CAD team [2].

8.3.1 Roll Bar

The first consideration in the design process for the chassis roll bar was that it had to extend 50 mm around driver's helmet in a normal driving position fastened in. On top of this, it must be metal and withstand a static load of 700N. The design for the roll bar can be seen in the 'Loading and Crash Protection' report [5] and it can withstand a maximum static load of 3.26 kN horizontally and 5.3 kN applied vertically. Buckling analysis was also performed and can be read in Devanshu Ghosh's report [5].

8.4 Tyres, Wheels, Axles and Hubs

The constraints governing the wheels and surrounding components were that the rims ought to be compatible with the dimension of the tyre for safety purposes, the wheel axle was to be designed for cantilever loads and also the location was specified. Wheels should be located inside body and must not touch the chassis or body or any other parts of the vehicle.

The design approach used to ensure compliance was maintained was to choose an existing BMX wheel and make any adjustments necessary.

8.5 Braking System

The Shell regulations surrounding the brakes were:

- 2 hydraulic braking systems – 1 per axle
- Single foot operated pedal controlling front wheel(s)
- Single lever action attached to steering wheel or single pedal
- Maximum 2 master cylinders
- Possible to activate both brakes without taking either hand off steering wheel [1]

On top of this, Imperial Racing Green further specified some of the pedal system was to be made from composites. The design achieved ultimately met these constraints and further information can be found in Pok Wei Hui's report [6].

8.6 Steering

The constraints surrounding the steering were largely sorted within the first 2 weeks of the project and these included:

- Front wheel steering only
- Turning radius of external wheel to follow 90 degree arc of 8m radius
- No contact between tyre and body or chassis
- Turning radius ≤ 8 m

Additional compliance discussions were had surrounding the choice of where to fix the cables from the steering column. Rules and regulations states that the floor of the vehicle must be solid i.e no gaps, thus it was discussed with Imperial Racing Green if a small slit could be placed in the floor to allow more compact positioning of cables. This was deemed acceptable as the ultimate 'floor' was the aerodynamics shell and this encompasses the entire vehicle. The design for this can be seen below, and more information can be found in Corey Taylor's report on steering [7].

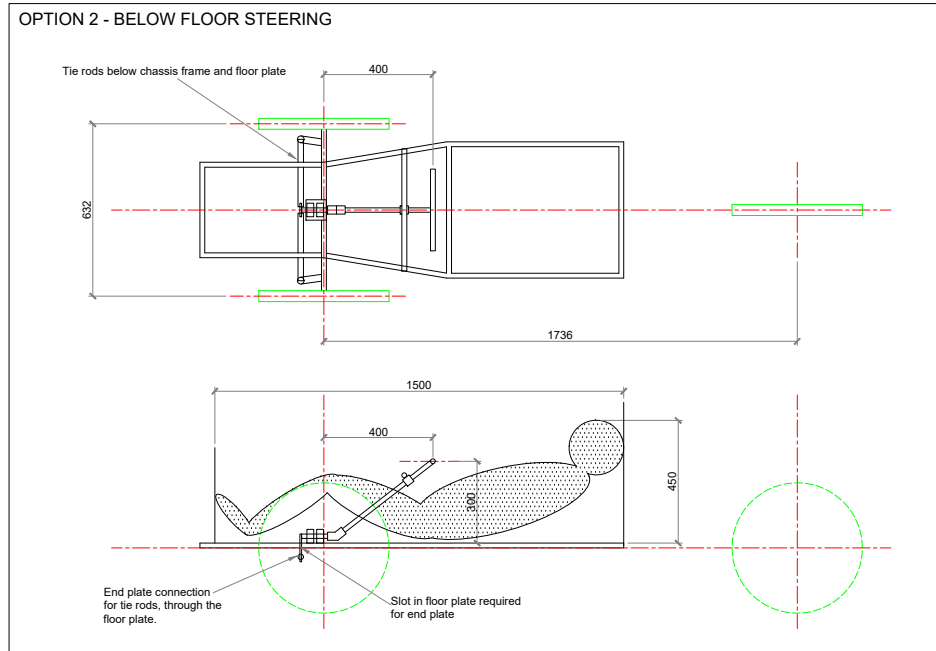


Figure 7: Drawing [7] showing the proposed location of the cables which was deemed compliant after discussions with compliance and Imperial Racing Green.

8.7 Motor

The constraints surrounding the motor were mainly related to the starter motor and the fact that guards were mandatory. The finalised design met these constraints and no issues were discovered at any point in the design process.

8.8 Driver Environment

The driver environment constraints involved the driver visibility, their comfort and the equipment available to the driver in terms of horn, hydration and clothing. These constraints were quickly met with little difficulty except for the horn placement as the horn has to be visible externally. Discussions were held with the aerodynamics, driver environment and compliance team to discuss this and it was decided to incorporate the ventilation system into this horn criteria. The design was deemed to be compliant despite its novel approach. The rear view mirrors were also aided by the driver environment team – there was a requirement for the mirrors to be $\geq 2500 \text{ mm}^2$ which was met. See the driver environment report for more detailed information on this [3].

8.9 Emergency Procedures

In the event of an emergency it is crucial that certain procedures are in place to maintain driver and spectator safety. These include shutdown mechanisms, both internal and external should it be required. The battery should be isolated from the electrics and a red push button must be clearly visible on the external of the car. These requirements were met by the design after multiple discussions with compliance and Imperial Racing Green and further details can be found in Tom Mrazek's report [8].

9 Additional Regulations

For the purpose of this project, the rules deemed ‘future issue’ as discussed earlier were not focused on. These should be taken into account by Imperial Racing Green in the next design phase after manufacturing. These rules included sticker placement on the vehicle, minimum driver competency standards and also off-track regulations. These, along with the spreadsheet created to manage rules will be handed back to Imperial Racing Green after this project has concluded to aid with their continued work.

In chapter 3 of the rules from Shell [1] there were also a number of constraints surrounding the event itself and the processes for the technical inspection and any subsequent disputes arising. These were not dealt with in the limitations of this project and will be a future issue for Imperial Racing Green.

10 Reflection

10.1 Resolved Compliance Issues

Throughout the project, regular meeting were held with every sub-team to guide the design process and any issues which arose from these meetings were dealt with accordingly. These included but were not limited to:

- Conceptual design – aerodynamics shell did not fit the chassis
 - This design was totally illegal as it firstly is not safe and secondly left no room for the required crumple zone
- A monocoque was discussed early on in the design process
 - This conflicted with the available budget from Imperial Racing Green
- The mirrors were proposed to be made utilizing cameras
 - On checking the rules no electronic substitute was acceptable so alternatives were found
- On final inspections for compliance the mirrors were found to be smaller than was required by the rules
 - This was flagged and changed accordingly
- The aerodynamics team desired to place the external shutdown behind a door to improve aerodynamic efficiency
 - This goes against the rule for the shutdown being easily accessible and visible
- The aerodynamics team desired a fully transparent top shell for visibility purposes
 - Whilst this was not against the rules per say, the driver could be potentially dangerously overheated which is unacceptable
- Rear wheel steering was desired early on in the project
 - This is explicitly against the rules outlined by Shell [1].
- The aerodynamics team designed the shell to have a sharp point at the rear for aerodynamic efficiency
 - This went against the regulation surrounding exposed sharp points so the design was rounded

10.2 Volume of Compliance Queries

The queries received were initially dealt with and researched by the compliance team and then if further clarification was needed it would be discussed with Imperial Racing Green or Shell. The graph below shows the number of queries throughout the project dealt with by compliance alone, joint discussions with Imperial Racing Green and also by contacting the organisers at Shell.

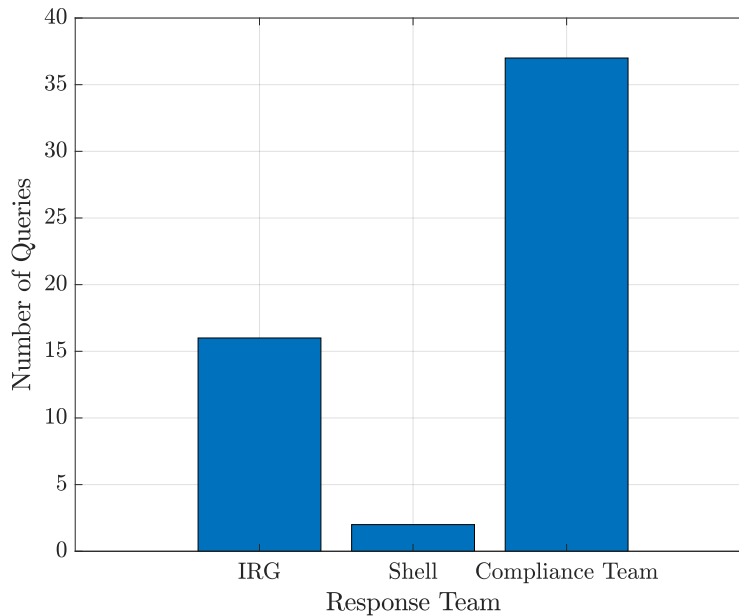


Figure 8: Graph highlighting the number of queries throughout the duration of the project and who helped resolve them.

10.3 Benefit of Separate Compliance Team

While some people could perceive it better to have compliance responsibilities held by the sub-teams solely, analysis on the response times of queries strongly suggests there is a significant time saving advantage by having a purposeful compliance team. The average time for each response option was recorded and can be seen in table 2. In addition the comparison of time taken to solve queries both with and without a dedicated compliance team can be seen in figure 9 – this was achieved by replacing the queries answered by the compliance team with the time taken for them to be answered by the next fastest alternative i.e Imperial Racing Green.

Response Team	Average time to respond	Number of queries	Total time expenditure
Shell	2880 minutes	2	5760 minutes
IRG and Compliance	118 minutes	16	1920 minutes
Compliance	15 minutes	37	555 minutes

Table 2: Table detailing the average length of time each response route took. This time was from the query being received to a definitive answer being returned.

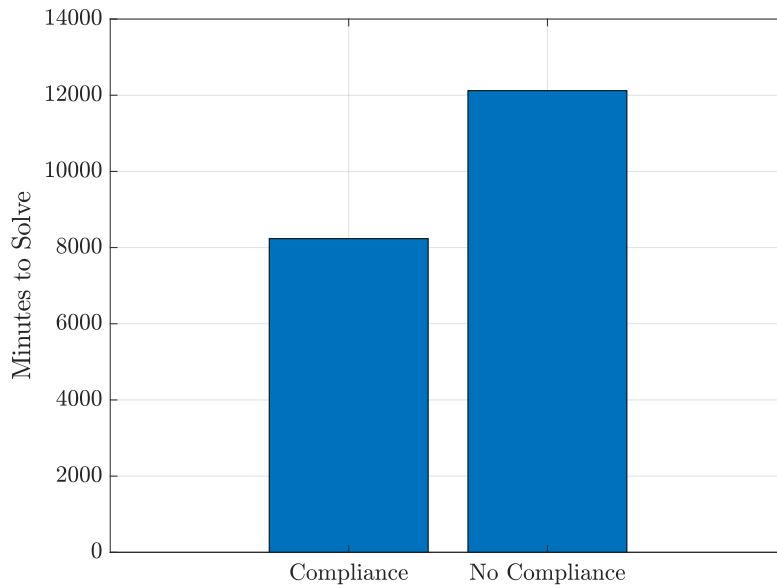


Figure 9: How a dedicated compliance team saved time during the project.

As can be seen from figure 9, the time saved by having a dedicated compliance team within the project saved 3885 minutes, which, over the 5 weeks of the project, equates to over 12 hours per week. In a project with a short time frame such as this one, 12 hours a week waiting for answers in order to progress with design decisions would be incredibly detrimental.

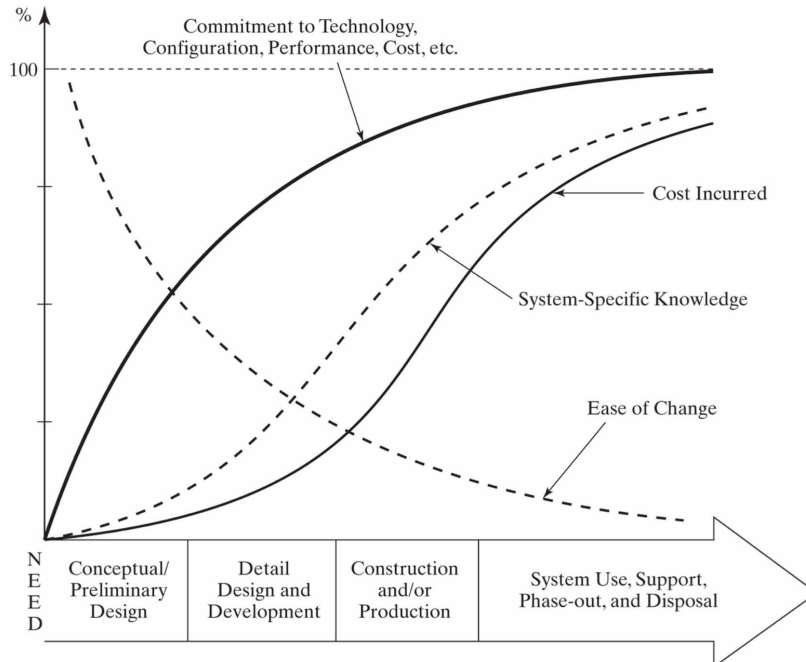


Figure 10: The benefit of effective compliance can be seen from the graph above [9]. By having consistent control of the rules from the project kickoff, major rework can be avoided at later stages. This is shown by the 'ease of change' line on the above graph – it is relatively easy to make changes at the conceptual design level.

10.4 Improvements

The process for compliance throughout this project was sound, efficient and effective. It utilised the experience of existing Imperial Racing Green members whilst allowing for new ideas to be explored. To further improve the process for compliance, one could ultimately explore looking at ways to find loopholes in these regulations that would allow for inventive unseen components or ideas. This type of strategy could see a team gain a significant advantage over competitors. However this could be potentially detrimental if the organisers identified and subsequently closed the loophole thus rendering the design illegal.

11 Conclusion

The compliance of the vehicle was demonstrated to be satisfactory against the aforementioned rules and regulations. It meets the constraints of both the budget and the safety requirements for Imperial Racing Green. Primary (with compliance team) and secondary (with Imperial Racing Green) checking of the rules ensured that no design mistake could be made throughout the process.

It should be noted that as this project was carried out entirely remotely, the challenges of ensuring compliance were magnified and therefore regular and precise communication was essential to ensure maximum results.

Regulatory Body	Regulation	Compliant?
Shell	Chapter 1 Rules and Regulations	Yes
IRG	Budget	Yes

Table 3: The table above shows the constraints were met, to add to this, the budget available was £12k and the design cost is £3712.71 which is well within the budget. As mentioned previously, the rules relevant to this stage of the design were met, with additional future concerns highlighted to Imperial Racing Green.

References

- [1] Shell. 2020. For Student Teams. [ONLINE] Available at:
<https://www.shell.com/make-the-future/shell-ecomarathon> [Accessed 12 June 2020].
- [2] Yang, Y., 2020. *Shell Eco Marathon – CAD*. London: Imperial College London.
- [3] Cole, M., 2020. *Shell Eco Marathon – Driver Environment and Interface*. London: Imperial College London.
- [4] Zijun, L., 2020. *Shell Eco Marathon – Packaging*. London: Imperial College London.
- [5] Ghosh, D., 2020. *Shell Eco Marathon – Loading and Crash Protection*. London: Imperial College London.
- [6] Wei Hui, P., 2020. *Shell Eco Marathon – Dynamics*. London: Imperial College London.
- [7] Taylor, C., 2020. *Shell Eco Marathon – Steering*. London: Imperial College London.
- [8] Mrazek, T., 2020. *Shell Eco Marathon – Drive Train*. London: Imperial College London.
- [9] Blanchard, B. and Fabrycky, W., 2013. *Systems Engineering and Analysis*. 5th ed. London: Pearson Education Limited.

A Organisational Layout

The organisational approach adopted by the team to ensure compliance was effectively managed throughout the duration of the project.

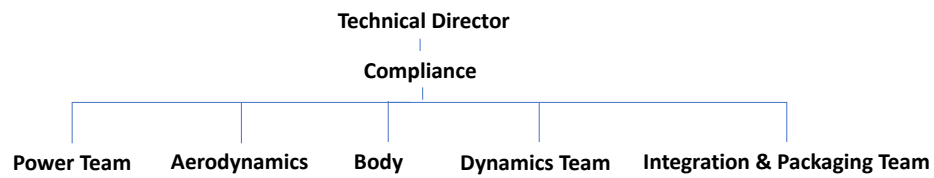


Figure 11: How the compliance team fit into the project organisation.

B Spreadsheet of Rules

The spreadsheet below demonstrates how rules were documented. (Note this is only a small segment of the spreadsheet).

Rule/Regulation Category	Constraint	Additional Info	Primary Sub-Team Concerned	Additional Sub-Team Concerned	Compliant?
Budget	Extends 50 mm around driver's helmet	In normal driving position fastened in. No visual impairment	Chassis	Driver Environment&Interface	Yes
Chassis roll bar	≥ 100 mm	Between front vehicle body & driver's feet	Chassis	Loading&Crash Protection	Yes
Min crumple zone	Solid floor and frame	No part of driver touches ground	Chassis	Aerodynamics	Yes
Body	Fully covered	No open top	Chassis		Yes
Vehicle max height	≤1000 mm	Measured between midpoints where tyre of reference wheel touch ground	Conceptual Design		Yes
Vehicle track width	≤125 mm		Conceptual Design		Yes
Height&track width	21000 mm		Conceptual Design		Yes
Vehicle wheelbase	51300 mm		Conceptual Design		Yes
Max vehicle width	53500 mm		Conceptual Design		Yes
Max total length	5140 kg	Without driver	Conceptual Design		Yes
Max weight	Direct arc of 90 deg	Inc full gear & comms. Ballast fitted if min weight not met	Driver Environment&Interface	Aerodynamics	Yes
Viability	350 kg		ING		Yes
Driver weight	Isolate and seal propulsion and energy storage from driver		Packaging	Driver Environment&Interface	Yes
Bulkhead	Prohibited		Aerodynamics		Yes
Aerodynamic appendages	≥ 50 mm or covered with foam/deformable material		Aerodynamics		Yes
Sharp points	Smoke, odour, sound	CONFIRM WHAT'S REASONABLE			Yes
Environmental considerations	Prohibited				Yes
Asbestos	Isolation of battery from electrics	Power controller not permitted. Relays must be normally op	Battery&Cooling	Power&Control	Yes
Emergency shutdown	Driver operated	At rear and installed on non-detachable part	Battery&Cooling	Power&Control	Yes
Internal shutdown	Means of latching red push button only reactivated by rotation	Push/pull lever not accepted	Battery&Cooling	Power&Control	Yes
Shutdown mechanism (external)	Electric battery		Battery&Cooling	Power&Control	Yes
Energy sources	60 volts	Inc. battery etc.	Battery&Cooling	Power&Control	Yes
Max voltage	1 allowed	No accessory battery allowed	Battery&Cooling	Power&Control	Yes
Battery	Outside driver's compartment behind bulkhead		Battery&Cooling	Power&Control	Yes
Battery location	Provide cell balancing and overvoltage protection during off-track charging		Battery&Cooling	Power&Control	Yes
Battery management system	Cell-level over-discharge, cell-level overcurrent and battery over-temperature to be provided in on-vehicle system		Battery&Cooling	Power&Control	Yes
BMS location	In physical battery package		Battery&Cooling	Power&Control	Yes
Self-built batteries	Constraints apply	CHECK IF SELF-BUILT	Battery&Cooling	Power&Control	Yes
Battery max capacity	1000 Wh		Battery&Cooling	Power&Control	Yes
Battery protection	Provided when charging. Battery enclosed in charging bag or solid metal containment tray under the battery.		Battery&Cooling	Power&Control	Yes

Figure 12: Spreadsheet of rule documentation.